

Mining High-quality Samples from Raw Data and Majority Voting Method for Multimodal Emotion Recognition

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ABSTRACT

Automatic emotion recognition has a wide range of applications in human-computer interaction. In this paper, we present our work in the Multimodal Emotion Recognition (MER) 2023, which contains three sub-challenges: MER-MULTI, MER-NOISE, and MER-SEMI. We first use a vanilla semi-supervised method to mine high quality samples from the MER-SEMI unlabeled dataset to expand the training set. Specifically, we ensemble three models trained with the official training set by a majority voting method, which is used to select samples with high prediction consistency. The selected samples together with the original training set are further augmented by adding noise. Then, the features of different modalities of expanded dataset are extracted from several pre-trained or fine-tuned models, and they are subsequently used to create different feature combinations to capture more effective emotion representations. Besides, we employ early fusion of different modal features and late fusion of different recognition models to obtain the final prediction. Experimental results show that our proposed method improves the performance over the official baselines by 30.4%, 55.3% and 1.57% for the three sub-challenges and ranks 4, 3, and 5, respectively. The present work sheds light on high-quality data mining and model ensemble by majority voting for multimodal emotion recognition.

CCS CONCEPTS

• **Computing methodologies** → **Artificial intelligence; Neural networks; Transfer learning.**

KEYWORDS

pre-trained models, fine-tune, multimodal emotion recognition

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1 INTRODUCTION

Affective computing is one of the most important parts of human-computer interaction, which aims to understand the emotional state of human beings [5, 9]. Human emotion expressions are usually reflected by multimodal information, such as facial expressions, intonation, language, and changes in body language [26], etc. Early studies usually use hand-crafted features for emotion recognition, such as pitch, energy, Mel Frequency Cepstrum Coefficient (MFCC), and DeepSpectrum [1] for audio modality and facial action units (FAUs) [18] for visual modality. With the development of deep learning techniques, many pre-trained models for audio, video and text have emerged, such as wav2vec2.0 [2], HuBERT [17], WavLM [7], BERT [11], MPNet [25] and MAE [15]. The pre-trained models are either directly used to extract embeddings [8, 19, 22], or fine-tuned with downstream tasks first and then extract more distinctive representations [6, 23]. In addition, to make full use of the complementary information of different modalities, researchers usually employ fusion methods for multimodal emotion recognition which include feature-level fusion (or early fusion), decision-level fusion (or late fusion), model-level fusion, and hybrid-level fusion [27].

The MER 2023 motivates global researchers to build innovative and robust technologies for multimodal emotion recognition [20]. It has three sub-challenges. The first one is MER-MULTI which aims to explore the relationships between discrete and dimensional emotions. It provides discrete and valence labels for training and test data, so we need to predict both of them. The second one is MER-NOISE which is designed to evaluate the robustness of multimodal emotion recognition systems in real scenarios. In particularly, the test data of both video and audio modalities are corrupted with noise using the OpenCV [4] and MUSAN [24]. The last one is MER-SEMI which provides a large amount of unlabeled training data. Its test data are also included in officially provided training data. However, to emphasize generalization ability of proposed methods rather than optimizing for a specific subset, the organizers do not provide the information about which samples belong to the test set.

In this paper, in order to make it easier and more efficient for the model to learn the emotion information from the audio modality pre-trained model, we first fine-tuned HuBERT using task-similar audio data, followed by extracting the hidden representation of HuBERT as the audio modality features. Considering that body language and other information are important for emotion recognition, we introduce representation of CLIP [28] model as one of the video modality features. We also employ the ResNet-FER2013 [16] model to extract facial features. Finally, the CLIP representation and facial features are concatenated together as the visual modality input of the multimodal model. In lexical modality, we use MacBERT [10]

to encode text. In addition, to improve the robustness of the model, we exploit a vanilla semi-supervised approach to data mining and data augmentation using noise addition. The experimental results show that both the feature combination and data augmentation approaches we used are very effective for multimodal emotion recognition.

2 MULTIMODAL FEATURE REPRESENTATION

In this section, we detail data pre-processing and feature extraction for each modality. The features of all modalities are first extracted at the frame/word-level and then aggregated into utterance/video-level features using average pooling.

2.1 Pre-processing

This challenge provides the dataset in the format of video clips (see Table 1 for details), so we need to pre-process the dataset. The discrete emotion labels are *neutral*, *anger*, *sadness*, *happiness*, *worry*, and *surprise*. Valence $\in [-5, 5]$ is the dimensional label. We process the dataset to obtain video, audio and text using the script provided by Baseline¹. Here, we use FunASR² [14] for transcribing speech to obtain text. In addition, we use the MUSAN and OpenCV to add noise to the train set and test3 to obtain the train-corrupt set and test3-corrupt set.

Table 1: The Details of MER 2023 Dataset

Partition	Name	# of samples label unlabeled	Duration
Train & Val	train set	3373 0	03:45:47
MER-MULTI	test1	411 0	00:28:09
MER-NOISE	test2	412 0	00:26:23
MER-SEMI	test3	834 73148	67:41:24

2.2 Feature Representation

Audio Features: We try two hand-crafted features: IS09 [21] and eGeMAPS [12]. All of them are extracted by openSMILE toolkit [13]. We also try neural network features extracted from pre-trained model HuBERT-large³ (HL). To explicitly bootstrap the multimodal model to learn the emotional features of the audio modality, we use the M3ED [29] dataset to fine-tune the Hubert-large model. Then, we extract the last hidden state of fine-tuned HuBERT-large model. We call this feature as FTHL.

Visual Features: We first extract the face image in the video using OpenFace toolkit [3], and use the Resnet-FER2013 (RF) model to extract the facial expression features. Then we use CLIP-base (CB) and CLIP-large⁴ (CL) model to encode the body language and other information in images. In order to utilize both face and body language features, we concatenate the Resnet-FER2013 features with the CLIP image features as CLIP-Face-large (CFL) features.

Lexical Features: We use MacBERT-large⁵ [10] (ML) to extract lexical features.

¹ <https://github.com/zeroQiaoba/MER2023-Baseline>

² <https://github.com/alibaba-damo-academy/FunASR>

³ <https://huggingface.co/TencentGameMate/chinese-hubert-large>

⁴ <https://huggingface.co/IDEA-CCNL/Taiyi-CLIP-Roberta-large-326M-Chinese>

⁵ <https://huggingface.co/hfl/chinese-macbert-large>

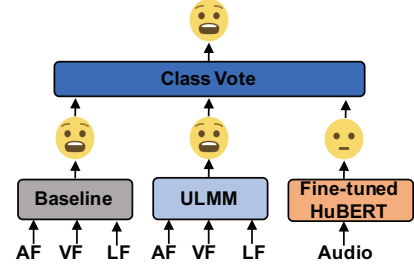


Figure 1: The method of the MER-SEMI sub-challenge. AF, VF, and LF represent audio features, visual features and lexical features respectively.

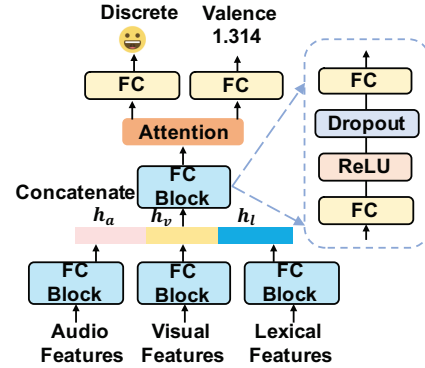


Figure 2: The framework of ULMM. FC denotes the fully connected layer.

Algorithm 1: The method of class voting in MER-SEMI

Input : C_a, C_b, C_c are discrete labels predicted by FT-HuBERT, ULMM, Baseline, $CL = [happiness, worry, surprise]$

Output: The result of voting: r_c

```

1 if  $C_b == C_c$  and  $C_b \neq C_a$  and  $C_b$  in  $CL$  then
2    $r_c = C_b$ 
3 else
4    $r_c = C_a$ 
5 end

```

3 METHOD

3.1 Data Augmentation

We will introduce the MER-SEMI first, since the other two sub-challenges will use the data mined from it. The method is shown in Figure 1. We use the official model of MER 2023 [20] as the Baseline. The utterance-level multimodal model (ULMM) and fine-tuned HuBERT (FT-HuBERT) model are our implementations. For details of ULMM, please refer to Section 3.2. The method of fine-tuning HuBERT is to add two fully connected layers on the top of the pre-trained HuBERT-large model for discrete and dimensional label prediction. The data used for fine-tuning are the training set of MER 2023.

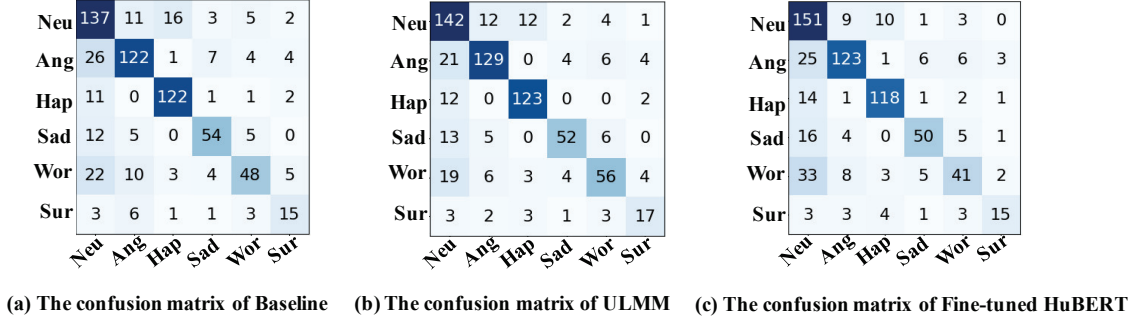


Figure 3: The confusion matrix for five-fold cross-validation of different models on the training set. Neu, Ang, Hap, Sad, Wor and Sur denote neutral, anger, happiness, sadness, worry and surprise respectively.

When we conduct five-fold cross-validation on the training set, we find an experimental conclusion through the confusion matrix (Figure 3) of the above modeling method. FT-HuBERT has higher accuracy in recognizing *neutral*, *anger*, and *sadness* samples, while multimodal Baseline and ULMM are better at distinguishing samples labeled as *happiness*, *worry*, and *surprise*. This is because the emotions like *worry* and *surprise* are more complex and require complementary information from other modalities to be better distinguished. Therefore, we design a class voting method (see Algorithm 1) that utilizes the advantages of multimodal model to compensate for the disadvantages of FT-HuBERT to get the final prediction.

In addition, we exploit a vanilla semi-supervised method to mine high-quality samples from the unlabeled dataset *test3*. Specifically, we first use the above three models trained with the official training set to predict the emotion labels for the *test3* and select the samples with consistent prediction results for Baseline [20], ULMM and fine-tuned Hubert. Then we use a confidence level of 0.99 to select 200, 200, 400 and 250 samples with pseudo-labels as *neutral*, *anger*, *sadness*, and *happiness*, respectively. Using a confidence level of 0.95 to select 241 and 165 pseudo-labeled samples of *worry* and *surprise*, respectively. The reason for the different confidence levels used and the resulting numbers of samples for each class is that the classes of the original dataset are unbalanced and *worry*, *surprise* is difficult to recognize. Finally, we obtain 1436 samples that are referred to as semi-dataset. By adding noise to these samples, we create another dataset, referred to as semi-dataset-corrupt.

3.2 Model Ensemble

In the MER-MULTI and MER-NOISE, the models we use are all ULMM. The structure of ULMM is the same as Baseline[20], but the parameters and features used for training are different, as shown in Figure 2. The features of each modality are first encoded by their respective FC Blocks, which consists of two fully connected layers, ReLU activation and Dropout layer. Then we concatenate the output of the FC Blocks to form the input of attention which plays the role of modality fusion. Finally, the model predicts the discrete label and valence value. The calculation method of attention is shown as follows:

$$h_i = \text{Concatenate}(h_i^a, h_i^v, h_i^l) \quad (1)$$

$$\alpha_i = \text{Softmax}(h_i^T W_\alpha + b_\alpha) \quad (2)$$

Algorithm 2: The majority vote for MER-MULTI or NOISE

Input : C_a, V_a are the prediction of ULMM trained with T, TC, S or SC. C_b, V_b are the prediction of ULMM trained with T, S or SC. C_c, V_c are the prediction of ULMM trained with T.

Output: The result of voting: r_c, r_v

```

1 if  $C_a == C_b == C_c$  then
2    $r_c = C_a; r_v = (V_a + V_b + V_c)/3$ 
3 else
4   if  $C_b == C_c$  and  $C_b \neq C_a$  then
5      $r_c = C_b; r_v = (V_b + V_c)/2$ 
6   else
7      $r_c = C_a; r_v = V_a$ 
8   end
9 end
```

$$z_i = h_i \alpha_i \quad (3)$$

where h_i^a, h_i^v, h_i^l represents audio, visual and lexical features of the i -th dimension. α_i denotes the modal attention weight and W_α, b_α are trainable parameters. z_i denotes the multimodal representation after attention fusion.

When evaluating the unimodal feature, the attention layer and the FC Blocks of other modalities in ULMM need to be removed. To further improve the performance, we design a majority voting method (see Algorithm 2). In this paper, all T, TC, S, and SC denote the training set, train-corrupt, semi-dataset, and semi-dataset-corrupt datasets, respectively.

4 EXPERIMENTAL SETUP

We use 256 and 128 units for the first and second FC layers of each FC Block (see Figure 2). We use a learning rate of $5e-4$, a dropout rate of 0.5, a batch size of 32, and the Adam optimizer for training the models. Fine-tuning HuBERT uses four RTX 3090 with a batch size of 1, and the gradient accumulation steps is 2. The optimizer is AdamW and the learning rate is $1e-5$. We use the cross-entropy loss as the emotion classification loss $\mathcal{L}_e$ and the mean squared error (MSE) as the valence regression loss $\mathcal{L}_v$. The joint loss $\mathcal{L}$ which optimizes all trainable parameters is defined as follow:

$$\mathcal{L} = \mathcal{L}_e + \mathcal{L}_v \quad (4)$$

Table 2: The performance of different feature combinations on ULMM, which was trained on training set. A, V, and L denote audio, visual, and lexical modalities, respectively.

Features			MULTI	NOISE	SEMI
A	V	L	Com	Com	WF1
Unimodal Results					
IS09	-	-	-0.491	-0.320	0.1774
eGeMAPS	-	-	-0.672	-0.486	0.0865
HL	-	-	0.353	0.163	0.8511
FTHL	-	-	0.482	0.259	0.8569
-	RF	-	0.258	0.229	0.5143
-	CB	-	0.301	0.255	0.5221
-	CL	-	0.386	0.420	0.5959
-	CFL	-	0.431	0.433	0.6050
-	-	ML	-0.06	-0.08	0.4632
Trimodal Results					
Baseline [20]			0.56	0.41	0.8640
FTHL	CFL	ML	0.623	0.597	0.8435

Evaluation Metrics: The evaluation metric of MER-SEMI is the weighted average F-score (WF1). The evaluation metric of MER-MULTI and MER-NOISE is defined as follow [20]:

$$Com = WF1 - 0.25 \times MSE \quad (5)$$

5 RESULTS AND DISCUSSION

5.1 Unimodal and Multimodal Comparison

Table 2 shows the performance of the three test sets on different modal features. The experimental results show that the performance of hand-crafted features is significantly inferior to that of the representation of pre-trained models. FTHL performance is much higher than HL, suggesting that similar tasks can lead downstream tasks to perform better.

CLIP is a multimodal pre-train model that mines their relations for the matched text and image data. It may also encode body language and facial expressions from visual modality. Therefore, it performs much better than RF. Our proposed CFL feature achieves the best performance in video modality. This is because face features and body language features can complement each other, thereby improving performance. The poor performance of the lexical features is due to the fact that the text of the dataset is usually very short and insufficient to express the appropriate emotions.

For audio modality, the performances for the NOISE sub-challenge are much worse than those for the MULTI sub-challenge, whereas for video modality the performance differences are narrowed and even the performances for the NOISE sub-challenge are slightly better. So the noise immunity of audio modality is not as good as that of video modality. Finally, we fuse the best-performing features of each modality to obtain the best experimental performance.

5.2 Data Augmentation and Model Ensemble

It can be seen from Table 3 that the performance of the model trained using both T and S is better than in the case where only T is used. This proves the effectiveness of our semi-supervised approach. Moreover, Experiments 3 and 4 demonstrate that our

noise-added data augmentation method improves the performance and robustness of the model. In addition, using Algorithm 2 to vote on the above results achieves the optimal performance.

Table 3: The performance of ULMM on MULTI and NOISE with different training data and voting methods.

Dataset					MULTI	NOISE
ID	T	TC	S	SC	Com	Com
1	✓				0.623	0.597
2	✓		✓		0.646	0.618
3	✓	✓	✓		0.6621	-
4	✓	✓		✓	-	0.628
Baseline [20]					0.56	0.41
Voting with ID 1, 2, 3					0.678	-
Voting with ID 1, 2, 4					-	0.637

In the MER-SEMI, due to the constrained time of the competition, we only complete the training of the model with the addition of semi-data. As shown in Table 4, the experimental results show an improvement over Baseline [20], ULMM, and FT-HuBERT performance. Finally, using Algorithm 1 for multi-model voting achieves the best performance.

Table 4: The performance on SEMI with different training data and voting methods.

Methods	SEMI (WF1)	
	T	T+S
Baseline [20]	0.8640	0.8662
ULMM	0.8435	0.8547
FT-HuBERT	0.8711	0.8727
Vote	0.8797	0.8811

6 CONCLUSIONS

In this paper, we present our solutions to the challenges of MER 2023. We first use a vanilla semi-supervised method to mine high quality samples from the MER-SEMI unlabeled dataset to expand the training set and use the method of adding noise for data augmentation. Then, the features of different modalities are extracted from several pre-trained or fine-tuned models. In addition, we also introduce the CFL features for enhancing the video modality representation. Moreover, we employ early fusion of different modal features and late fusion of different recognition models to improve the recognition performance. Experimental results show that our proposed methods improve the recognition performance and robustness.

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